

Question number	Answer	Additional guidance	Mark
8(a)	An answer that includes one of the following points: • correct reason for benefit of optimal hydration (1)	there will be other answers e.g.to regulate body temperature keep joints lubricated (1) prevent infections / reduce inflammation (1) deliver nutrients to cells, (1) keep cells / tissues/organs functioning properly (1) enable {metabolic/ enzyme / chemical} reactions to occur (1) maintain structure of cells maintain salt concentration / water potential / allow osmosis to occur decreases ageing IGNORE prevent dehydration / digestion unless linked to hydrolysis	(1)

Question number	Answer	Additional guidance	Mark
8(b)	An explanation that includes three of the following points: • caffeine (and its metabolites) absorbed into the blood (1) • (carried to brain) where it crosses the blood brain barrier (1) • where it binds to (specific) receptors (on neurones) (1) • caffeine causes {sodium ion channels open /allows neurones to continue {transmission / depolarisation} / more action potentials / release of neurotransmitters} (1)	ACCEPT blocks adenosine receptors (in synapse) ACCEPT caffeine enhances {the effects of dopamine / increases {cortisol/ dopamine} secretion}	(3)

Question number	Answer	Additional guidance	Mark
8(c)	An answer that includes the following points: • correct suitable reason (1) • correct relevant explanation of reason (1)	• larger surface area • larger (muscle) mass • more cells • males have a greater energy requirement • males have less fat • males have higher metabolic rate • so more water lost by (evaporation) • more heat released so more sweating to maintain body temperature • more water needed in {enzymic reactions / respiration} • as muscles store less water than fat • women store more fat than men so can have a bigger store of water in fat than men there may be other acceptable responses	(2)

Question number	Answer	Additional guidance	Mark
8(d)	An explanation that includes five of the following points: • by homeostasis / osmoregulation (1) • osmoreceptors {in the brain / hypothalamus} detect + (changes in) water potential (of blood) (1) • impulses sent to the pituitary gland (1) • releasing {more / less} ADH (1) • causing the {DCT / collecting duct} to reabsorb {more / less} water/ sodium ions (1)	ACCEPT reference to process of sodium concentration being too high or too low IGNORE concentration ACCEPT releasing more / less aldosterone ACCEPT causing {DCT / collecting duct} to become {more / less} permeable / {more / fewer} aquaporins in {DCT / collecting duct}	(5)

Question number	Answer	Additional guidance	Mark
8(e)	An explanation that includes the following points: • water moves out (from the cell) by osmosis (1) • because concentration of solutes is higher {outside the cell / in blood / in tissue fluid} (than in the cell) / (1)	ACCEPT higher water potential inside cell / lower water potential {outside the cell / in blood / in tissue fluid} accept water moves down {water potential / water concentration} gradient	(2)

Question number	Answer	Additional guidance	Mark
8(f)	An answer that includes two of the following points: • {increased / high} levels of nitrate in the blood (1) • results in relaxation of smooth muscle (of blood vessels) (1) • leading to {vasodilation / widening of lumen} (1)	ACCEPT {nitrite / NO / nitrous oxide / nitric oxide / metabolite of nitrate} for nitrate ACCEPT nitrate transported to heart ACCEPT stimulation of SAN to give more depolarisation ACCEPT leading to increased {stroke volume / cardiac output / heart rate}	(2)

Question number	Answer	Additional guidance	Mark
8(g)	A description that includes three of the following points: • compare {groups / individuals} consuming [energy drinks v non energy drinks/ before and after drinking energy drinks} (1) • perform ECG (1) • {analyse / compare} ECG for (irregular) heart rhythm (1)	ACCEPT energy drink drinker and non-energy drink drinker ACCEPT descriptions e.g. interval between {same / named} peak changes ACCEPT compare to see the difference in ECG traces	(3)

Question number	Answer	Additional guidance	Mark
8(h)	An answer that includes the one of the following points: • correct reason (1) • explanation of reason (1)	Reason and explanation MUST be linked. e.g. an allergy to milk protein / lactose intolerance / cannot breakdown {lactose/ milk sugar} e.g. {inability to produce / deficiency} in enzyme lactase / {gene / allele} coding for lactase is missing /immune response to milk protein	(2)

